Self-Applicable Eye Strain Detection through the Measurement of Blink Rate Using Raspberry Pi

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***Abstract* — You Only Look Once (YOLO) is a Convolutional Neural Network (CNN) used primarily for object detection models due to its high accuracy and reliable processing speed. In the study, YOLO was used to calculate the blink rate of an individual through a twenty (20) second recording. The model utilized YOLOv5, which was trained using one hundred (100) photos of faces obtained from the public dataset CelebFaces Attributes (CelebA). The system used a Raspberry Pi 3B+ and a 5MP Camera Module for video recording. Testing the model on 40 respondents (20 seconds each with 160 frames) yielded an overall accuracy of 85% when assessed using a confusion matrix. The paper benefits individuals who spend an increasing amount of time in front of a computer screen by helping them evaluate the stress level of their eyes. The results of the study can be used for future monitoring devices of eye strain.**

Keywords — YOLOv5, Raspberry Pi 3B+, eye strain, eye detection, machine learning

I. INTRODUCTION

The increase in accessibility of semiconductor devices brought by the growing influence of technology can be both beneficial and harmful to individuals. This advent of technology has caused people, especially children, to become more and more focused on their devices, increasing ocular conditions such as eye strain [1]. Additionally, the wavelengths of wireless signals used by mobile devices and wireless terminals (~2cm to 2.5cm) can lead to the early formation of cataracts [2]. The amount of time spent using electronic devices will lead to a higher probability of the surfacing of different symptoms, such as headaches, irritation of the eyes, and partial loss of vision, which can all be categorized under fatigue or tiredness. Previous studies also suggest that over 50% of all technology users have some form of digital eye strain [3]. Although eye strain can be caused by extended exposure to electronic devices, it can also be caused by exposure to dry air, fatigue, reading for an extended period, and even long-distance driving. The symptoms of digital eye strain are divided into two types: those related to binocular vision stress, which is the difficulty of focusing on an object with both eyes, and symptoms that would lead to dry eyes. Eye strain can also lead to symptoms such as nausea, pains in the back, neck, and shoulders, dry eyes, and watery eyes—blurred vision, eyelids twitching, drowsiness, and double vision [4][5]. Detection of digital eye strain consists of questionnaires as well as evaluations, which include a Critical Flicker-Fusion (CFF) frequency, which is the frequency where a flickering light is then perceived continuously by a person, blink rate, which is the number of blinks a person does per minute, and finally, various pupil characteristics that would see if the pupil contained the effects of visual fatigue. Blink rate is considered one of the significant symptoms of eye strain because of the altered blinking pattern, which would lead to a reduced blink rate and incomplete blinking [6]. The average blink rate of a person is between fifteen (15) to twenty (20) blinks per minute [7][8], but this is significantly reduced by up to five-fold when eye strain is present. The average blink rate of an individual under the effects of eye strain becomes three (3) to eleven (11) blinks per minute. The prototype will also make use of a Raspberry Pi 3B+ as the primary processing unit. This specific model of Raspberry Pi was used over the latest model due to their similarity in benchmarks, with their main difference being Random Access Memory (RAM) [9]. After comparing the specifications of the Raspberry Pi 3B+ model to the Raspberry Pi 4, the researchers were able to conclude that the 3B+ was the more cost-efficient option.

A study attempted to use blink rate and eye sclera to detect eye strain among children. They proposed using modified Otsu color tracking and achieved an accuracy of 83% [10]. To calculate an individual's blink rate, a system should first be able to locate the eyes to count each blink within a sixty (60) second frame. One of the methods for finding the eyes is using the Eye Aspect Ratio (EAR) equation. This equation is a metric for the openness or closedness of the eyes by using the Euclidean vertical and horizontal distances between the eyelids [11]. A study explored using EAR when implemented on a Raspberry Pi 3 microcontroller to capture live images of drivers to determine drowsiness [12]. Object recognition is a technique used in image processing that identifies objects in digital images or videos. The You Only Look Once (YOLO) algorithm is one example of an algorithm specializing in object recognition by encapsulating regions of interest of an image inside bounding boxes and classifying each one. In 2021, YOLO was used to classify Filipino foods with 77% accuracy [13]. Another study utilized the YOLO algorithm to track and analyze a driver's eyes to assess fatigue with an accuracy of 95%, 86%, and 89% for each tested driver [14]. A separate study attempted to use a Convolutional Neural Network (CNN) model trained to detect facial cues and achieved an accuracy of 77% [15]. Additionally, different research used eye images to train a model to recognize the iris and achieved a 93.34% success rate [16]. To accurately determine an individual's blink rate, the system should be able to track the eyes. A study utilized a lumination algorithm, which is an algorithm that uses light to measure objects and gather information regarding gaze, attention level, and fatigue levels [17]. Another study used a device called the Tobii Eye Tracker, which uses cameras, projectors, and infrared light to find the gaze point of an individual [18]. Additionally, a study aimed to improve a segmentation algorithm created by Ayra

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Panganiban to segment ideal and non-ideal iris images. Ideal iris images can be defined by their fixed dimensions, where the iris is in the center of the picture. On the other hand, non-ideal iris images can be off-angle or use a different size of iris; these characteristics can cause difficulty with iris recognition [19]. This study and other studies regarding object recognition use the Raspberry Pi microcontroller to run and control different electronic devices. One example of its utilization is implementing a camera module to classify four (4) types of Cactaceae Plants [20]. Another study used the microcontroller to identify diseases that could be found in different Oranda goldfish [21]. Furthermore, a Raspberry Pi was used to load the images of cells to identify the Anemia cells [22]. Moreover, implementing the YOLO algorithm on a Raspberry Pi module created an automatic inventory system with an accuracy of 89.55% [23]. Finally, another study used Raspberry Pi to train a model to identify if there is a snail in an image or not [24].

Currently, the existing methods of detecting eye strain can be difficult as they require specialized equipment and a trained professional to read the results. This study aims to explore a method using the YOLO algorithm to help provide a more accessible and cost-effective eye strain detection method.

This study aims to implement a prototype to determine if an individual suffers from eye strain using a face detection algorithm using YOLO to calculate the blink rate. This objective can be accomplished through the following: (a) to create a system using Raspberry Pi that can count the blinks per minute of an individual, (b) to utilize the YOLO Algorithm to be able to process the live video feed to detect blinks, and (c) to evaluate the accuracy of the system by comparing its output to the diagnosis of an ophthalmologist.

The significance of this study is to provide people, especially in the tech industry, with a way of monitoring their eye strain levels to avoid harmful effects. This paper also benefits children as they are becoming more literate when it comes to the use of technology, thus increasing their exposure to mobile devices and wireless signals. The device will benefit these individuals due to their extensive periods of exposure to electronic devices, which may result in eye strain. These individuals using this device could take the necessary steps to alleviate eye strain if found to have it.

The paper is limited to the detection of eye strain only. Other eye features, such as iris size, sclera size, and iris color, will not be included in the study. The study will be based on the results of previous studies regarding the correlation between blink rate and eye strain.

II. Methodology

A. Conceptual Framework

Fig. 1 shows the researchers' conceptual framework. The input comprises a twenty-second video recording. Then, the process will be the application of the YOLOv5 Algorithm Implementation. The algorithm implementation includes the trained facial recognition model to detect blinks and the eye strain assessment calculations. This research outputs the calculated number of blinks per minute of the individual and the eye strain assessment.

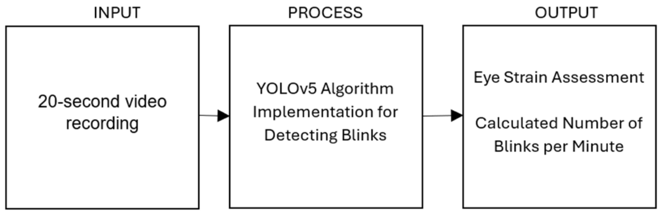


Fig. 1. Conceptual Framework.

The person will have eye strain if the prototype detects that the person has a blink rate of fifteen (15) to twenty (20) blinks per minute. Anything below or above, then the person has strained eyes. The final output will be an LCD screen showing the final blink rate and the predicted diagnosis of whether the person has eye strain.

B. Hardware Design

Fig. 2 shows the hardware block diagram the research used. It shows the different components that the researchers used to create the prototype. There are three inputs for the created prototype, as seen in the diagram. The first is the 5V DC power supply, which will power the Raspberry Pi 3 Model B+ and, in turn, the rest of the components. The following input is the Raspberry Pi camera module, which will capture a twenty-second video of the person's face. The last input is the button starting the program on the Raspberry Pi. The microcontroller to be used is the Raspberry Pi 3 Model B+, which will process the images and determine whether the person has eye strain. The output will be a 16x2 LCD showing the blink rate, along with the predicted diagnosis, and an LED, which will turn on upon button press to serve as a signal that the recording has started. The LED will then turn off after twenty seconds when the recording is finished.

The prototype that will be used for gathering data and testing will make use of a Raspberry Pi 3B+ model. This model was chosen over the Raspberry Pi 4 model because of its cost-effectiveness. The primary difference between the 3B+ model and the 4 model is their processing speed. The researchers have assessed that the detection model that will be used for the prototype does not demand a high-performing microcontroller. In addition to this, the Raspberry Pi 4 model also consumes a higher amount of power and generates a higher degree of heat, which can affect the lifespan of the other components in its vicinity. Lastly, the Raspberry Pi 3B+ model is also cheaper than the latest model 4.

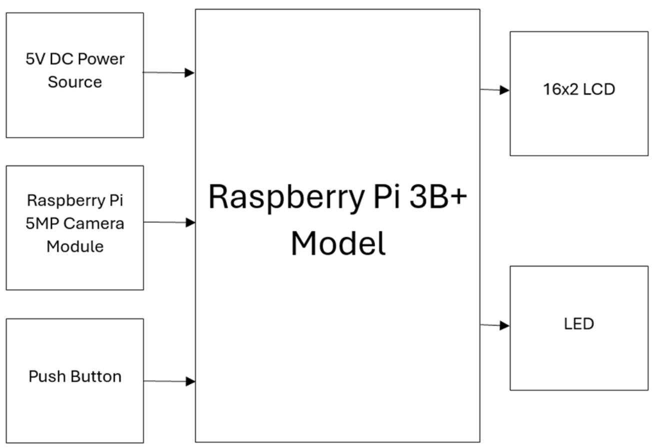


Fig. 2. Hardware Block Diagram.

C. Software Development

Fig. 3 shows the software flowchart of the main program, which shows the process the program will follow. First, the program will load the different dependencies that it needs. This includes the camera, the activation of the various Raspberry Pi pins, and the loading of the eye strain detection model. Once initialization is done, the user will be prompted to press the button to start the main program, otherwise nothing will happen. Upon pressing the button, the camera will start recording a twenty-second video of the individual. After the recording is finished, the video will undergo image processing to identify the area of the eyes. Then, the video will undergo image processing and feature extraction to determine the area of the eyes on the individual's face. Once finished, the results of the program will be displayed on the LCD module.

Fig. 4 shows the YOLOv5 Algorithm Implementation module, which is responsible for processing the video frame by frame to detect blinks. Each frame will be scanned by the model to create a bounding box surrounding the area of the eyes. Then, the model will attempt to assess each frame and classify them as either 'open' or 'closed'. If the model detects more than five consecutive frames labeled as 'closed', then the blink counter will increment by one. Once each frame of the twenty-second video has been processed, the blink counter will be multiplied by three to calculate the blinks per minute of the individual. This is required as the unit of measurement used for the eye strain assessment is blinks per minute. Once the blink rate calculations are done, the model will assess if the resulting blink rate value falls inside the range of the set threshold of fifteen to twenty blinks per minute, which is the average blink rate of an unstrained eye. Finally, the program will output the calculated blinks per minute of the person and the assessment of the model regarding their eye strain.

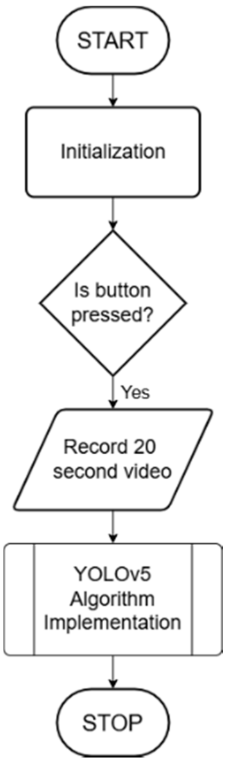


Fig. 3. Main Software Flowchart.

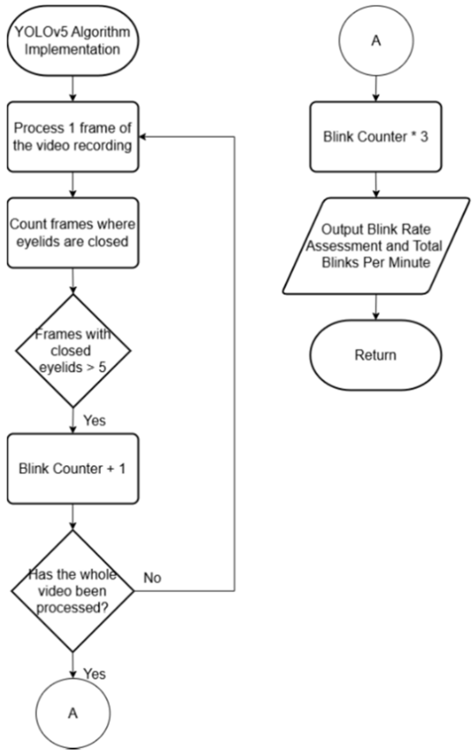


Fig.4. YOLOv5 Algorithm Implementation Module

D. Experimental Setup

Fig. 5 shows the front view of the prototype that will be used for data gathering. The figure shows the green LED light, 16x2 LCD module, camera module, and the push button located on the outside of the wooden casing. The camera module that will be used to record the twenty-second video sits on top of the casing to make it easier for the user to record themselves for the recording. Other components such as the Raspberry Pi 3B+ model and a Printed Circuit Board (PCB) can be located on the inside. A PCB was used instead of a breadboard to ensure the robustness of the connections, especially when relocating the prototype.

Before the user can start the program, they must first position themselves approximately one foot away from the camera module. This is to make sure that the camera records the whole face of the individual, which is vital to the face detection aspect of the model. It is also important that the background of the user does not contain other people, as their blinks can also be counted, thus resulting in an inaccurate prediction.

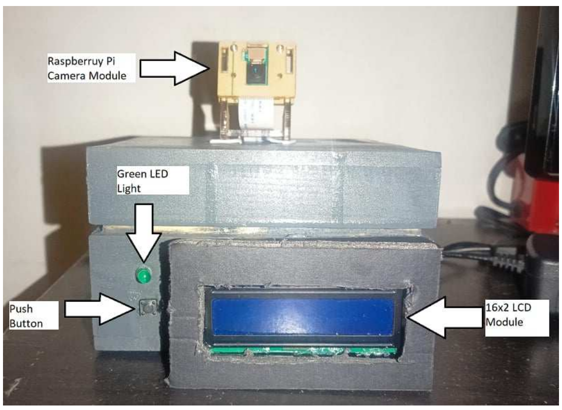


Fig 5. Front View of the Prototype.

E. Data Gathering

Before testing, the researchers trained a YOLOv5 model to classify if a person has open or closed eyes. One hundred images (100) were gathered from the CelebFaces Attributes (CelebA) dataset, which are pictures of the faces of different celebrities. These images were then given annotations for the training.

The researchers gathered forty (40) respondents to test the system. The prototype will take a video of their face for twenty (20) seconds, and right after, the system will use image processing on the recorded video to determine the blink rate and the predicted diagnosis of the system. An ophthalmologist will then diagnose the respondent to compare if the system can accurately predict if the individual suffers from eye strain.

III. Results and Discussion

A. Data Gathered

Table 1 shows a sample image of the algorithm classifying when eyes are either open or closed, which will be used to classify if there is a blink. A bounding box is also created around the area of the user's eyes to aid the assessment of eye strain.

Table II shows the testing table that was used in the research as well as the results that were conducted throughout the 40 trials. The table shows the trial number, the number of actual blinks per minute, the predicted blinks per minute, the doctor's diagnosis, and the predicted diagnosis. An online consultation takes the diagnosis from an ophthalmologist.

TABLE I. Sample Output from the Algorithm

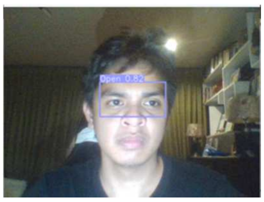
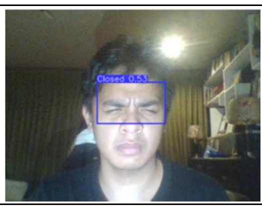
State	Output from the Prototype
Open Eyes	
Closed Eyes	

TABLE II. Testing Table

Trial	Blinks per minute	Predicted Blinks per minute	Doctor's Diagnosis	Predicted Diagnosis
1	27	26	Strained	Strained
2	15	17	Normal	Normal
3	33	38	Strained	Strained
4	30	31	Strained	Strained
5	30	30	Strained	Strained
6	15	12	Normal	Strained
7	27	25	Strained	Strained
8	42	42	Strained	Strained
9	36	37	Strained	Strained
10	21	21	Strained	Strained
11	27	25	Strained	Strained
12	15	14	Normal	Strained
13	18	19	Normal	Normal
14	18	16	Normal	Normal
15	15	16	Normal	Normal
16	19.5	18	Strained	Normal
17	21	21	Strained	Strained
18	0	0	Strained	Strained
19	6	5	Normal	Strained
20	0	0	Strained	Strained
21	21	25	Strained	Strained
22	12	21	Strained	Strained
23	12	12	Strained	Strained
24	12	11	Strained	Strained
25	12	15	Strained	Strained
26	12	5	Strained	Strained
27	12	15	Normal	Normal
28	21	21	Strained	Strained
29	30	25	Strained	Strained
30	21	23	Strained	Strained
31	27	28	Strained	Strained
32	36	45	Strained	Strained
33	39	44	Strained	Strained
34	39	54	Strained	Strained
35	12	224	Strained	Strained
36	12	11	Strained	Strained
37	9	7	Strained	Strained
38	18	13	Normal	Strained
39	27	35	Normal	Strained
40	15	15	Normal	Normal

B. Statistical Treatment

A confusion matrix was used to determine the system's accuracy. Each prediction will be classified under four categories: False Positive (FP), False Negative (FN), True Negative (TN), and True Positive (TP). The definition of different classifications are as follows: FP for a person without eye strain to be predicted to have eye strain, FN for a strained person predicted to not have eye strain, TN for a person that is correctly given the prediction of unstrained, and TP for a person that is correctly given the prediction of strained eyes. Given these distinctions, the following confusion matrix is created based on the results in Table II.

TABLE III. CONFUSION MATRIX

	Prediction Of Model		
Doctors Diagnosis		***Strained***	***Normal***
	Strained	30	1
	Normal	5	4

Equation (1) is then used to calculate the system's accuracy.

$$ACC = \frac{\sum_{n=1}^{2} A_{nn}}{\sum_{\substack{i=1 \\ j=1}}^{2} A_{ij}} \quad (1)$$

The model's accuracy can be determined by dividing the total number of correct strained and normal assessments made by the model by the total number of tests conducted. Using equation (1) above, the calculated overall accuracy of the system is 85%.

IV. CONCLUSION AND RECOMMENDATION

The study successfully created a program with the help of machine learning to accurately count the blinks per minute of an individual to detect eye strain. The prototype used a Raspberry Pi 3B+, its corresponding camera module, and a 16x2 LCD. The model's results were successful as it could accurately predict a doctor's diagnosis regarding eye strain with a percentage of 85%, which is an improvement on previous studies that were able to garner an accuracy of 83% using other algorithms. The researchers recommend that a better camera be used, an application be developed to make it easier to interact with the device, and a different algorithm be used. Using a higher version of the Raspberry Pi or a Jetson Nano is also possible for faster inference time and possibly better results. Additionally, future studies can also diversify the dataset used for training the model, as the current model has difficulty accurately counting the blinks of more pronounced cases of monolid eyes.

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